

Alert Investigation Report

Platform: SOC Simulation Project — WDLabs

Environment: Microsoft Defender for Endpoint / Microsoft Sentinel

Case Details

Field	Value
Alert Name	Hands-on keyboard attack was launched from a compromised account (attack disruption)
Alert ID / Link	152 / https://security.microsoft.com/incident2/152/overview?tid=567ea528-4e82-40fa-914e-38c05fd69677
Analyst	Carl
Date Investigated	8:09PM AEST 13/09/2026
Severity	High

1. Alert Summary

- **Host / Account affected:** socsim-dc.socsim.local and socsim-srv01.socsim.local
- **Date & Time of alert:** 31 Aug 2026 14:51:53
- **What triggered the alert:** Mimikatz credential theft tool
- **Why it triggered (rule/logic):** A malicious file masquerading as svchost.exe

2. Investigation Findings

What happened?

- 1 Aug 2026, 14:51 — Attacker attempted lateral movement over RPC from socsim-dc.socsim.local using the dave.brown account; blocked on multiple devices.
- 31 Aug 2026, 19:26 — Password of an account was changed; this same account later performed malicious activity, suggesting the attacker already had enough access to

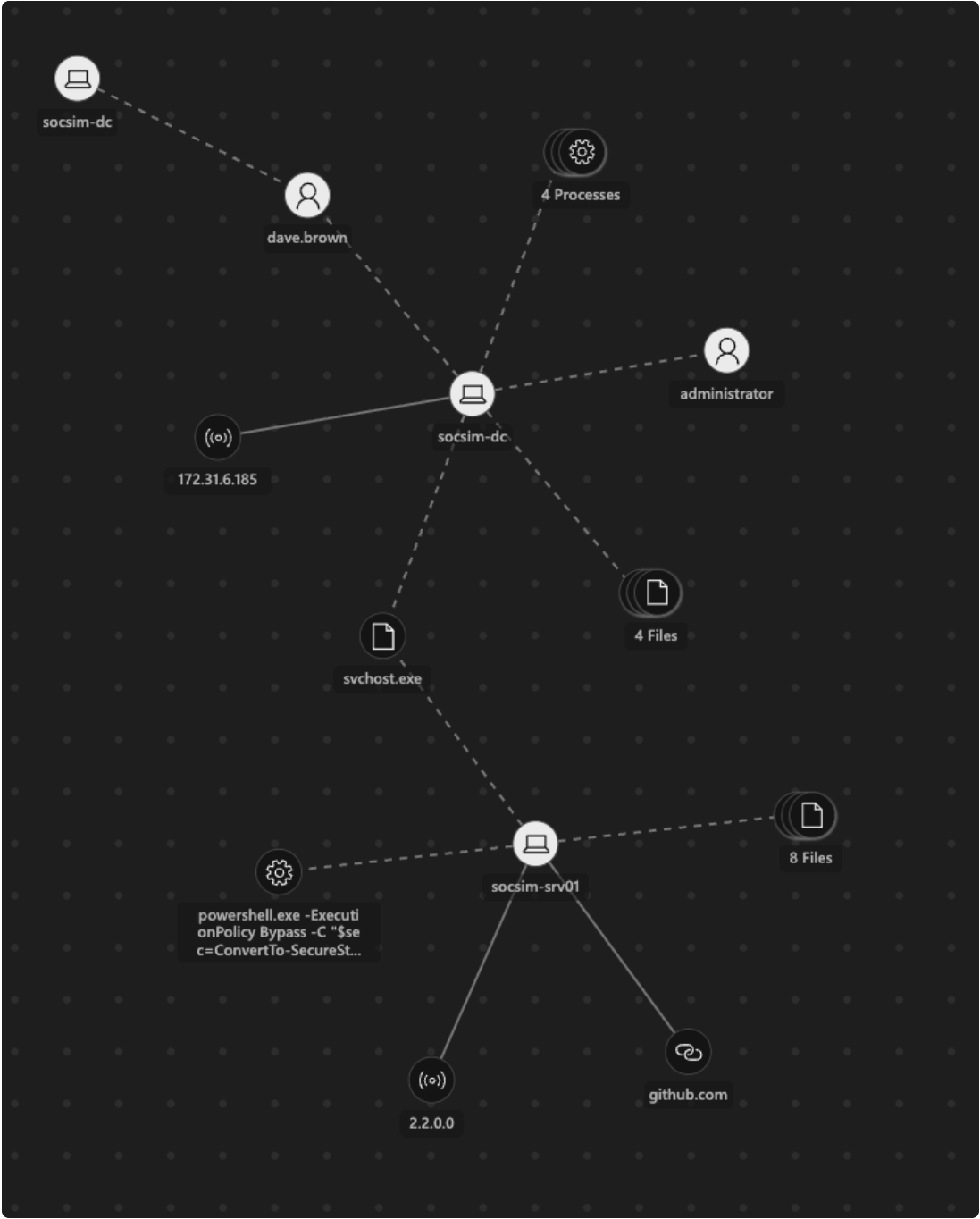
manipulate credentials.

- 1 Sep 2026, 12:00 — Mimikatz credential theft tool detected on socsim-dc.socsim.local under dave.brown, marking the start of active credential harvesting from memory.
- 1 Sep 2026, 12:37 — Defender flagged potential human-operated malicious activity, confirming a live attacker rather than automated malware.
- 1 Sep 2026, 12:45 — Five alerts fired almost simultaneously on socsim-dc.socsim.local: compromised account conducting hands-on-keyboard attack, repeated human-operated malicious activity alerts, and malware detected during lateral movement.
- 1 Sep 2026, 13:02 — Sensitive credential memory read detected, indicating an active attempt to pull credentials from protected memory.
- 1 Sep 2026, 13:03 — Peak of the attack, with a dense burst of correlated alerts across socsim-dc.socsim.local and socsim-srv01.socsim.local:
 - Suspicious remote activity
 - Another human-operated malicious activity alert
 - Mimikatz detected again, this time on socsim-srv01 (lateral spread to a second server)
 - Suspicious PowerShell command execution
 - Malicious credential theft tool execution confirmed
 - System file masquerade (fake svchost.exe)
 - Suspicious WMI process creation (common remote-execution technique)
 - Another compromised-account hands-on-keyboard alert
 - Suspicious access to the LSASS service (blocked)
 - A second confirmed credential theft tool execution on socsim-dc.socsim.local
- 1 Sep 2026, 13:37 — Attacker attempted RPC-based lateral movement again across multiple devices, showing continued attempts to spread even after detection.

Evidence

```
// Paste key log entries, process trees, or notable findings here
```

Process Tree



File Logs

Export

15 itemsCustomize columns

First seen	File name	Verdict	Remediation status	File path	Device
☐ 1 Sep 2026 13:04	m.zip	Malicious		C:\Windows\Temp\m.zip	socsim-srv0
☐ 1 Sep 2026 13:04	m.zip	Malicious		C:\Windows\Temp\m.zip	socsim-srv0
☐ 1 Sep 2026 13:04	svchost.exe	Suspicious		C:\Windows\Temp\svc...	socsim-dc
☐ 1 Sep 2026 13:03	hid.dll	Suspicious		C:\Windows\System32\...	socsim-dc
☐ 1 Sep 2026 13:03	samlib.dll	Suspicious		C:\Windows\System32\...	socsim-dc
☐ 1 Sep 2026 13:03	cryptdll.dll	Suspicious		C:\Windows\System32\...	socsim-dc
☐ 1 Sep 2026 13:03	WinSCard.dll	Suspicious		C:\Windows\System32\...	socsim-dc
☐ 1 Sep 2026 13:03	mimispool.dll	Malicious		c:\windows\temp\mk\x...	socsim-srv0
☐ 1 Sep 2026 13:03	mimilib.dll	Malicious		c:\windows\temp\mk\x...	socsim-srv0
☐ 1 Sep 2026 13:03	mimikatz.exe	Malicious		c:\windows\temp\mk\x...	socsim-srv0
☐ 1 Sep 2026 13:03	mimidrv.sys	Malicious		c:\windows\temp\mk\x...	socsim-srv0

Process Logs

Export

5 itemsCustomize columns

First seen	Process name	Verdict	Remediation status	Process ID	Device
☐ 1 Sep 2026 13:04	svchost.exe	Suspicious		1864	socsim-dc
☐ 1 Sep 2026 13:04	lsass.exe	Suspicious		840	socsim-dc
☐ 1 Sep 2026 13:03	ntoskrnl.exe	Suspicious		4	socsim-dc
☐ 1 Sep 2026 13:03	WmiPrvSE.exe	Suspicious		1172	socsim-dc
☐ 1 Sep 2026 13:03	powershell.exe	Suspicious		2296	socsim-srv0

IP address Logs

First seen	Address	Verdict
☐ 1 Sep 2026 13:04	172.31.6.185	Suspicious
☐ 1 Sep 2026 13:03	2.2.0.0	Suspicious

URL Logs

First seen		Address	Verdict	Remediation status
☐	1 Sep 2026 13:03	https://github.com/gentilkiwi/mimikatz/releases/dow...	Suspicious	

Activity Logs

Activity started	Title	Type	Category	Target	Performed by
☐ 6 Sep 2026 13:11	dave.brown uncontained	Uncontain User	Response	dave.brown	Attack Disruption
☐ 1 Sep 2026 19:46	Alert 'Lateral movement on RPC blocked... Alert link		Correlations		Microsoft Defender XDR
☐ 1 Sep 2026 13:44	Alert 'Lateral movement on RPC blocked... Alert link		Correlations		Microsoft Defender XDR
☐ 1 Sep 2026 13:17	Stop and Quarantine File	Quarantine File	Response	mimilib.dll	AIR
☐ 1 Sep 2026 13:14	Alert 'Compromised account conducting... Alert link		Correlations		Microsoft Defender XDR
☐ 1 Sep 2026 13:14	Alert 'Potential human-operated malicio... Alert link		Correlations		Microsoft Defender XDR
☐ 1 Sep 2026 13:13	Alert 'A user changed the password of a... Alert link		Correlations		Microsoft Defender XDR
☐ 1 Sep 2026 13:11	Alert 'Potential human-operated malicio... Alert link		Correlations		Microsoft Defender XDR
☐ 1 Sep 2026 13:10	Alert 'Potential human-operated malicio... Alert link		Correlations		Microsoft Defender XDR

Alerts

Attack story

Alerts (23)

Activities (31)

Assets (4)

Investigations (2)

Evidence and Response (23)

Summary

↓

Export

📅

6 Months

▼

🔔

Report incident inaccuracy

19 Alerts

🔍

Search for name or ID

⚙️

Customize columns

Filter set:

Status: New, In progress

✕

🔍

Add filter

🔄

Reset all

☐ Alert name ▼	Tags ▼	Severity ▼	Investigation state ▼	Status ▼	Category ▼	Impacted assets ▼
☐ Lateral movement on RPC blocked on multi...	Attack Disruption	Informational		New	Lateral Movement	🖨️ socsim-dc 👤 dave ⋮
☐ Potential human-operated malicious activity		High		New	Suspicious activity	🖨️ socsim-dc 👤 dave ⋮
☐ Compromised account conducting hands-o...		High	Queued	New	Lateral Movement	🖨️ socsim-dc 👤 dave ⋮
☐ Suspicious WMI process creation		Medium		New	Execution, Discover...	🖨️ socsim-dc 👤 dave ⋮
☐ System file masquerade		Medium		New	Defense Evasion	🖨️ socsim-dc 👤 dave ⋮
☐ Malicious credential theft tool execution de...		High		New	Credential Access	🖨️ socsim-dc 👤 dave ⋮
☐ Potential human-operated malicious activity		High		New	Execution, Lateral ...	🖨️ socsim-dc 👤 dave ⋮
☐ Potential human-operated malicious activity		High		New	Execution, Lateral ...	🖨️ socsim-dc 👤 dave ⋮
☐ Sensitive credential memory read		High		New	Lateral Movement, ...	🖨️ socsim-dc 👤 dave ⋮
☐ Mimikatz credential theft tool		High	Partially Remediated	New	Credential Access	🖨️ socsim-srv01 👤 ⋮

Was the activity successful?

- ☒ Yes
- ☐ No
- ☐ Unknown

Verdict

- ☒ **Malicious**
- ☐ **Non-malicious**
- ☐ **Suspicious — requires further investigation**

Justification:

This conclusion is based on direct evidence, not just the alert names. The Evidence tab showed the actual Mimikatz files (mimikatz.exe, mimilib.dll, mimispool.dll, mimidrv.sys) sitting on disk on socsim-srv01, confirming the tool was really present. The Process list showed lsass.exe itself flagged suspicious at the same timestamp — LSASS is the exact memory Mimikatz targets, proving credential theft was actually attempted. A URL entity pointed straight to the tool's real GitHub source (gentilkiwi/mimikatz), showing where it was downloaded from. The incident graph tied all of this to one account (dave.brown) moving from socsim-dc to socsim-srv01 via a disguised svchost.exe file and a PowerShell command. Finally, the Activities log showed Microsoft Defender's Attack Disruption engine independently took real response actions — quarantining the malicious file and containing the account — confirming this was treated as a genuine active compromise, not a false positive.

3. Investigation Steps

1. The attack story - looking the tree process was easy to look at and understand
2. The alerts told on how the attack story occurred
3. Opening up the investigate tab

Queries Used

```
// Paste queries here
```

```
SecurityAlert  
| where TimeGenerated > ago(30d)  
| where AlertSeverity == "High"  
| project TimeGenerated, AlertName, AlertSeverity, Description, CompromisedEntity, Tactics  
| sort by TimeGenerated desc
```

4. Categorisation

- ☒ **True Positive** — malicious activity confirmed
 - ☐ **Benign True Positive** — alert fired correctly, activity is legitimate
 - ☐ **False Positive** — alert fired incorrectly, rule needs review
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5. Recommended Actions

Escalation:

- ☐ Urgent — phone call required
- ☐ Non-urgent — email sufficient
- ☐ No escalation required

Containment / Next Steps:

- Call the user to change passwords and perform windows defender virus scanning
- Call a meeting with seniors and other experts on this matter when stuck
- Recommend to always perform windows update when possible
- Change all admin and service account passwords. The attacker stole credentials from the domain controller, so we have to assume they have them all.
- Reset the krbtgt password twice. This stops the attacker from creating fake login tickets that would let them back in.
- Lock the dave.brown account again. It was unlocked on 6 Sep and is currently active.
- Check that dave.brown's password was really changed. The attacker may have set it themselves on 31 Aug.
- Finish cleaning socsim-srv01. The Mimikatz alert still says only partly fixed.
- Delete the leftover bad files: mimikatz.exe, mimispool.dll, mimidrv.sys, m.zip, and the fake svchost.exe.
- Look closely at mimidrv.sys. It's a driver file, which means the attacker may have had deep access to the system.
- Rebuild both socsim-dc and socsim-srv01 instead of just cleaning them.
- Find out how the attacker got in. The alerts start on 31 Aug when the attack was already happening, so the break-in was earlier.
- Check dave.brown's login history from at least a week before 31 Aug. Look for strange IP addresses or failed logins.
- Check if 172.31.6.185 is a normal computer on the network or something unknown.
- Block internet downloads on servers. A domain controller was able to download Mimikatz straight from GitHub, which should not happen.

- Check what the administrator account was doing. It shows up in the incident graph, so it may have been used too.
- Assign this incident to someone and set a classification. 19 of the 23 alerts are still marked as New.

Alert Tuning:

☐ Tune recommended —

☒ No tuning required

6. Lessons Learned

- looking at the process graph too long
 - could've done more deep diving on the logs
 - only looked the logs as a quick brief
 - looking at the investigation page for too long
 - used AI for too long to conclude investigation - more practice and reps on investigating incidents quick and use AI when the incident is too confusing to recognize
 - Didn't use much KQL command much - manually browsing the alerts
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►  About this template